Inhwa Song • Statement of Purpose

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My research objective lies at the intersection of human-computer interaction (HCI), social computing, and AI. Many challenges at both individual and societal levels arise from the limitations of personal worldviews and perspectives. I have witnessed how such limitations can contribute to prejudice and division at a societal level through my experience of serving as a student president during the COVID crisis. At an individual level, these limitations often result in struggles with mental health, knowledge I learned while training as a peer counselor. Advances in social and language technologies present opportunities to empower individuals and communities to expand their worldviews. AI agents, for instance, have the potential to simulate dialogues of viewpoints different from the user. Moreover, online platforms have increased global connectivity. However, these technologies are often designed and developed in ways that not only fail to realize this potential but sometimes exacerbate existing issues—algorithms can trap users in echo chambers, while AI systems may reinforce personal ideology or overlook marginalized perspectives. Through my research, I aim to **harness the potential of social and language technologies, responsibly, to empower individuals and communities to actively expand their worldviews, fostering both societal harmony and personal growth.** To this end, I hope to pursue a Ph.D. in Information at Michigan University.

My interest has led me to many exciting research projects across diverse academic domains and industry settings. As a peer counselor, I witnessed firsthand how individuals project and shape their worldviews during personal challenges. Some shared their experiences of using LLM chatbots, highlighting both potential and risks. Motivated by these insights, I seized the opportunity to collaborate with **Prof. Munmun De Choudhury** during my exchange period at Georgia Tech. I led timely research to examine how individuals understand and use LLM chatbots when seeking support for their mental health needs in real-world settings [2]. I conducted semi-structured interviews with 21 participants from every inhabited continent by connecting with local forums and support groups. Grounding analysis in psychotherapy literature of what makes support effective, we introduced the concept of therapeutic alignment. Our work has been successfully submitted to *CSCW'24* with me as the first author, currently undergoing review following a request for revisions. I realized that providing interaction components for the user to gain control within a reliable structure is the key design consideration of building responsible generative AI systems, further motivating me to not only envision but to build and test the reliability of such systems.

This motivation led me to a research internship at NAVER AI Lab to work with **Dr. Young-Ho Kim**, whom I

have long admired for his novel system contributions to HCI research. I developed ExploreSelf, an LLM-driven interactive system that supports users in exploring their narratives of personal challenges by actively engaging with adaptive guidance [1], grounding in a formative study. Even with my vast hands-on experience managing the end-to-end development and deployment of complex computing systems, this particular work was especially valuable to me for three main reasons. First, I gained a deeper understanding of what system contribution means in HCI research. Second, this research I led received a *Revise & Resubmit decision in CHI'25* with compliments for the high research quality. Last but not least, I recognized my ability to quickly learn and apply new technologies and programming methods, along with the confidence that I can effectively tackle any new technical challenges I face in future research.

During my internship at NAVER AI Lab, I could communicate with AI researchers from diverse backgrounds, each bringing a different worldview. For instance, within the goal of responsible AI, stakeholders' perspectives diverged between creating safety benchmarks, embedding safety in LLMs, and investigating what responsible AI means in the context of Korean society. Through direct conversations with stakeholders, I saw firsthand how such divergent views stemmed from the tension between qualitative societal contexts and the quantitative methods that dominate ML. These discussions reinforced my commitment to promoting the expansion of worldviews. They also motivated me to broaden my own perspective on navigating the balance between technical feasibility and social requirements. While I have experienced translating qualitative insights into concrete system design [2], I also wanted to tackle research questions that involve direct intervention in language models. Under the guidance of **Prof. Alice Oh**, my ongoing project involves constructing a dialogue dataset of unstructured learner inputs paired with concise queries by fine-tuning the Phi-3 model with the StackExchange dataset, where each post combines a body of long-form explanations and a title of a focused question.

These research projects gave me the opportunity to adopt a wide range of approaches, from system building and evaluation [1,8], large-scale data analysis [4], fine-tuning language models, to qualitative methods such as semi-structured interviews [2] and cultural probe [3], and also systematic reviews [7]. Above all, I have developed my ability to identify valuable research questions and learned how to transform an initial idea into systematic research.

My goal is to become an HCI researcher to inform, design, and build social computing systems that foster social interaction with mutual understanding by respecting each other's worldviews. I strongly believe that the best way to achieve this is to pursue an Information Ph.D. program at the University of Michigan due to its interdisciplinary and collaborative research environment. In particular, **Professor Eytan Adar** and **Professor Eric Gilbert** parallel

my interests in mutual agreement across individuals interacting in social computing systems, such as in terms of consent, and content sharing.

Additionally, I believe that the interdisciplinary nature of HCI highlights the importance of collaboration among individuals from various backgrounds. The University of Michigan's vibrant community and the Information School's interdisciplinary nature will offer the necessary support for my academic success. Once I finish my doctoral degree, my goal is to become a professor who promotes inclusive collaboration. I strive to contribute to academia and society by connecting research and industry through practical studies and mentoring future students.

References:

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